Software Requirements Specification (SRS)

Project: Electronics Store Inventory Management System

Phase 1 — Requirements / Validation Specification

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1.0	09-09-2026	Team H9	Initial SRS and validation specification for electronics inventory management	

Approvals

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12. Phase 1 Rubric Compliance Summary

The following table maps the Phase 1 deliverable requirements provided by the course rubric to the corresponding sections of this document. Existing SRS material is retained; this summary identifies where each required deliverable is addressed.

Phase 1 Requirement	Document Location
1. Problem statement analysis / feasibility testing	Sections 1.5 and 2.6
2. SRS List	Sections 4, 5 and 5.1 contain the functional, non-functional and security requirements.
3. Categorize requirements into FRs and VFRs	Section 4 contains ESIM-F requirements; Section 9 contains ESIM-VFR requirements mapped to the FRs.
4. Prepare RTM table for the requirements	Section 8 — Requirements Traceability Matrix.
5. Identify the number and names of actors	Section 7.2 explicitly identifies 4 actors / external roles.
6. Group the use cases based on actors	Section 7.2 provides actor-wise grouping.
7. Draw Use Case Diagram on Actors	Section 7.1 contains the two existing UML use-case diagrams.
Penetration testing — planning only	Section 5.1.3 provides the planned penetration-testing scope and validation cases; no Phase 1 execution results are claimed.

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the requirements for an Electronics Store Inventory Management System developed using Python and Django. The system maintains component inventory records, supports add/update/delete operations, shows availability and low-stock information, and provides reports and a dashboard for inventory monitoring.

1.2 Scope

The system covers authentication, component registration, inventory viewing, searching/filtering, stock updates, component deletion, availability calculation, low-stock identification, dashboard indicators, inventory reports, input validation, authorization, and persistent database storage.

The initial release excludes customer-facing online purchasing, payment processing, supplier payment management, full accounting, and advanced demand forecasting.

1.3 Audience

The document is intended for the project team, course coordinator, project guide, developers, testers, project evaluators, store administrators, store staff, and managers who interact with or evaluate the system.

1.4 Definitions

1.5 Problem Statement Analysis

The Electronics Store Inventory Management System is proposed to address the difficulty of maintaining and monitoring electronic component inventory in an organized and reliable manner. An electronics store may have many components with different identifiers, categories, prices, stock quantities and minimum stock levels. Without a centralized inventory system, checking availability, identifying low-stock items, updating records and preparing inventory reports can become difficult and error-prone.

The proposed system provides a centralized web-based solution for component registration, inventory viewing, searching and filtering, stock updates, deletion or deactivation, availability checking, low-stock identification, dashboard monitoring and inventory reporting. Authentication, authorization, validation and persistent storage are included to support controlled and reliable inventory management.

Problem Statement Summary

The problem is therefore defined as the need for a centralized, role-controlled and testable system that maintains accurate electronics inventory records and provides timely information about component availability, stock levels and inventory status.

Term	Definition
Component	An electronic part or product maintained as an inventory item.
Inventory	The collection of component records and their current stock quantities.
Component ID	A unique identifier used to distinguish an inventory item.
Minimum Stock Level	The threshold at or below which an item is considered low stock.
Availability	The current stock status of a component based on the defined quantity rule.
CRUD	Create, Read, Update and Delete operations.
RTM	Requirements Traceability Matrix linking requirements to design/module references and tests.
FR	Functional Requirement — a function the system shall provide.

Term	Definition
NFR	Non-Functional Requirement — a measurable quality or constraint requirement.
VFR	Validation/Verification Requirement — a measurable criterion used to verify a requirement.

2. Overall Description

2.1 Product Perspective

The Electronics Store Inventory Management System is a web-based Django application. Users access the system through a browser. Django handles application logic, authentication, validation and database interaction. A relational database stores component, stock, user and reporting data.

2.2 Major Product Functions

- User login and logout with role-based authorization.
- Add new electronic components with required inventory details.
- View all stored components in an organized inventory list.
- Search and filter components by identifier, name or category.
- Update component details and stock quantities.
- Delete or deactivate an unwanted component after confirmation.
- Calculate and display component availability.
- Identify components at or below the minimum stock level.
- Display inventory totals and low-stock indicators on a dashboard.
- Generate inventory reports for monitoring and decision-making.
- Validate input values and display clear error messages.
- Persist valid inventory data in a database.
- Maintain basic change information for inventory updates where supported.

2.3 User Roles and Characteristics

Role	Characteristics / Responsibilities
Inventory Administrator	Maintains inventory and manages permitted users/roles; can perform inventory-changing operations.
Store Staff	Checks inventory, availability and reports; performs only operations allowed by assigned permissions.
Store Manager	Uses dashboard, availability, low-stock information and inventory reports for monitoring and decisions.
QA / Evaluator	Validates the implementation against requirements and planned acceptance tests.

2.4 Operating Environment

Client: modern web browser such as Chrome, Edge or Firefox. Server/application: Python runtime with Django on Windows, Linux or macOS. Database: a relational database supported by Django, such as SQLite for development or PostgreSQL/MySQL for larger deployment. Development tools may include VS Code, Git and a browser-based test environment.

2.5 Constraints

2.6 Feasibility Testing

Technical Feasibility

The proposed system is technically feasible because Python and Django provide the required web-application framework, authentication, validation and database-access capabilities. Django ORM can connect the application to a relational database. The project can use SQLite for development, with PostgreSQL or MySQL as alternatives for larger deployment. The required client environment is a modern web browser.

Operational Feasibility

The system is operationally feasible because inventory administrators, store staff and store managers can interact with the application through a browser. The functions are aligned with the responsibilities described in Section 2.3. No dedicated hardware is required for normal operation.

Economic Feasibility

The system is economically feasible for the academic project because Python, Django and commonly used development tools are open-source or freely available. Development can be performed using existing student hardware and software resources without specialized infrastructure.

Schedule Feasibility

The project is schedule-feasible because the initial release is limited to core inventory-management functionality. Customer-facing online purchasing, payment processing, supplier payment management, full accounting and advanced demand forecasting are explicitly outside the initial scope.

- The project is limited by the academic development schedule and available student resources.
- The initial release focuses on core inventory management rather than complete enterprise ERP functionality.
- Only authorized users shall perform inventory-changing operations.
- Component identifiers shall be unique within the inventory.
- Stock quantities and monetary values shall use valid non-negative values.
- The system depends on availability of the application server and database.

3. External Interface Requirements

3.1 User Interfaces

The web interface shall provide a login page, dashboard, inventory list, component form, search/filter controls, availability indicators, report interface, user/role controls where enabled, confirmation prompts for destructive actions, and clear success/error messages.

3.2 Hardware Interfaces

- No dedicated hardware is required.
- Users require a display and keyboard/mouse or equivalent input device.
- The application server requires a machine capable of running Python/Django and the selected database.

3.3 Software Interfaces

- Python runtime and Django web framework.
- Django ORM for database access.
- Relational database supported by Django.
- Modern web browser for the user interface.
- HTML/CSS and JavaScript for the client-side interface where required.

3.4 Communications

- Client and server shall communicate using HTTP/HTTPS.
- HTTPS shall be preferred for deployed or non-local environments.
- Form submissions and API requests, where used, shall use validated request data.
- Authentication/session information shall be protected and shall not be exposed through ordinary application messages or logs.

4. System Features (Detailed)

Each functional requirement is written in a testable form and includes type, priority, source/stakeholder, acceptance criterion/test reference and dependencies. IDs follow ESIM-F-###.

4.1 User Authentication

Description: The system shall authenticate an existing user using valid credentials before granting access to protected inventory functions.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
ESIM-F-001	The system shall authenticate an existing user using valid credentials before granting access to protected inventory functions.	Functional	High	User / Admin	AC: valid credentials create an authorized session; invalid credentials are rejected. Test: TC-AUTH-01	Django authentication

4.2 Logout & Session Management

Description: The system shall allow an authenticated user to log out and shall invalidate the associated session.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
ESIM-F-002	The system shall allow an authenticated user to log out and shall invalidate the associated session.	Functional	High	User	AC: after logout, protected pages/actions cannot be accessed without re-authentication. Test: TC-AUTH-02	Session management

4.3 Add Component

Description: The system shall allow an authorized user to create a component record containing a unique ID, name, category, description, unit price, quantity and minimum stock level.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
ESIM-F-003	The system shall allow an authorized user to create a component record containing a unique ID, name, category, description, unit price, quantity and minimum stock level.	Functional	High	Inventory Administrator	AC: one valid submission creates one record. Test: TC-INV-01	Component database

4.4 View Inventory

Description: The system shall display stored component records with relevant inventory details and current availability.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
ESIM-F-004	The system shall display stored component records with relevant inventory details and current availability.	Functional	High	Staff / Manager	AC: stored test records are displayed correctly. Test: TC-INV-02	Database

4.5 Search / Filter

Description: The system shall allow users to search or filter inventory by component ID, name and/or category.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
ESIM-F-005	The system shall allow users to search or filter inventory by component ID, name and/or category.	Functional	Medium	Staff / Manager	AC: results contain only matching records. Test: TC-SEARCH-01	Inventory list

4.6 Update Component

Description: The system shall allow an authorized user to update permitted component details and stock quantity.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
ESIM-F-006	The system shall allow an authorized user to update permitted component details and stock quantity.	Functional	High	Inventory Administrator	AC: edited value persists and no duplicate record is created. Test: TC-INV-03	Existing component

4.7 Delete Component

Description: The system shall allow an authorized user to delete/deactivate a component after confirmation.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
ESIM-F-007	The system shall allow an authorized user to delete/deactivate a component after confirmation.	Functional	High	Inventory Administrator	AC: confirmed deletion removes the record from active inventory. Test: TC-INV-04	Authorization / database

4.8 Availability Status

Description: The system shall calculate and display availability using current quantity and the defined minimum-stock rule.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
ESIM-F-008	The system shall calculate and display availability using current quantity and the defined minimum-stock rule.	Functional	High	Staff / Manager	AC: status matches the documented quantity rule at boundary values. Test: TC-AVAIL-01	Quantity / threshold

4.9 Low-Stock Identification

Description: The system shall identify components whose quantity is at or below the minimum stock level.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
ESIM-F-009	The system shall identify components whose quantity is at or below the minimum stock level.	Functional	High	Staff / Manager	AC: all qualifying test records are identified. Test: TC-LOW-01	Availability rule

4.10 Dashboard

Description: The system shall display dashboard indicators such as total components, available stock and low-stock items.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
ESIM-F-010	The system shall display dashboard indicators such as total components,	Functional	Medium	Manager / Admin	AC: dashboard totals match the test database. Test: TC-DASH-01	Inventory data

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
	available stock and low-stock items.					

4.11 Inventory Reports

Description: The system shall generate inventory reports containing relevant component, quantity and availability information.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
ESIM-F-011	The system shall generate inventory reports containing relevant component, quantity and availability information.	Functional	Medium	Manager / Admin	AC: report data matches stored test records. Test: TC-REP-01	Reporting data

4.12 Input Validation

Description: The system shall reject missing mandatory fields, duplicate component IDs, negative quantities, invalid prices and other invalid input.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
ESIM-F-012	The system shall reject missing mandatory fields, duplicate component IDs, negative quantities, invalid prices and other invalid input.	Functional	High	QA / User	AC: invalid data is rejected with a clear message. Test: TC-VAL-01	Validation rules

4.13 Persistent Storage

Description: The system shall store valid inventory records in a database and retrieve them across user sessions.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
ESIM-F-013	The system shall store valid inventory records in a database and retrieve them across user sessions.	Functional	High	System	AC: stored record remains available after logout/login. Test: TC-DATA-01	Database

4.14 Authorization

Description: The system shall restrict add, update, delete and user-management operations to users with required permissions.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
ESIM-F-014	The system shall restrict add, update, delete and user-management operations to users with required permissions.	Functional	High	Admin / Security	AC: unauthorized attempts are blocked. Test: TC-SEC-01	Authentication / roles

4.15 Error Handling

Description: The system shall display understandable error messages when an operation fails or invalid information is submitted.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
ESIM-F-015	The system shall display understandable error messages when an operation fails or invalid information is submitted.	Functional	Medium	QA / User	AC: error identifies the failed/invalid operation and valid data remains intact. Test: TC-ERR-01	Exception handling

4.16 Inventory Change Tracking

Description: Where implemented, the system shall retain basic information about inventory changes such as the changed item, action and time of change.

Req ID	Requirement (shall...)	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case ref	Comments / Dependencies
ESIM-F-016	Where implemented, the system shall retain basic information about inventory changes such as the changed item, action and time of change.	Functional	Low	Admin / QA	AC: supported update creates the defined change record. Test: TC-AUD-01	Audit log

5. Non-Functional Requirements (Detailed)

NFRs below are measurable and tied to planned test cases. IDs: ESIM-NF-###.

Req ID	Requirement (shall...)	Category	Priority	Acceptance criteria / Measurement
ESIM-NF-001	For normal academic test load, 90% of inventory list, search and CRUD requests shall complete within 2 seconds.	Performance	High	TC-PERF-01
ESIM-NF-002	The system shall support at least 1000 component records in the project test database without functional failure.	Scalability	Medium	TC-PERF-02
ESIM-NF-003	A valid inventory transaction shall either complete successfully or leave the affected record in its previous valid state.	Reliability	High	TC-REL-01
ESIM-NF-004	During a controlled test run, the application shall remain available for at least 99% of the scheduled test duration, excluding planned maintenance.	Availability	Medium	TC-REL-02
ESIM-NF-005	A new user shall be able to log in, view inventory and perform permitted common tasks without developer assistance; actions shall provide clear feedback.	Usability	Medium	TC-UX-01
ESIM-NF-006	The Django application shall separate presentation, business logic, authentication/authorization and data-access responsibilities into identifiable modules.	Maintainability	Medium	Code review
ESIM-NF-007	The application shall function correctly on current versions of Chrome, Edge and Firefox.	Compatibility	Medium	TC-COMP-01
ESIM-NF-008	A database backup of the project inventory shall be restorable in the defined development environment.	Recoverability	Medium	TC-REC-01

5.1 Security

5.1.1 Security Objectives

- Protect inventory and user information from unauthorized access.
- Ensure only authenticated and authorized users can change inventory data.
- Preserve integrity of component quantities and master records.
- Prevent common invalid-input and web-application attacks through validation and framework controls.

5.1.2 Security Requirements

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case
ESIM-SR-001	Protected application pages shall require authentication.	Security	High	TC-SEC-01
ESIM-SR-002	The system shall enforce	Security	High	TC-SEC-02

Req ID	Requirement (shall...)	Type	Priority	Acceptance criteria / Test case
	authorization for add, update, delete and user-management operations.			
ESIM-SR-003	Passwords shall not be stored in plaintext; Django's supported password hashing mechanism shall be used.	Security	High	TC-SEC-03
ESIM-SR-004	The application shall use CSRF protection for applicable state-changing web requests.	Security	High	TC-SEC-04
ESIM-SR-005	User input shall be validated on the server before database-changing operations are performed.	Security	High	TC-SEC-05
ESIM-SR-006	Application logs shall not record passwords or authentication secrets.	Security	High	TC-SEC-06

5.1.3 Security Validation Plan

Security procedures are defined in the Software Test Plan (STP), Section 5.1.

6. Quality Attributes & Acceptance Tests

The functional, performance, security, usability and compatibility acceptance suites and exit criteria are defined in the Software Test Plan. Planned test cases are listed in the Test Cases document. Results are recorded in the Requirements Traceability Matrix and the Phase 3 test evidence.

7. System Models and Diagrams

The following UML use-case diagrams show the external actors and principal system interactions for inventory/component management and reporting/availability/monitoring. The architecture and sequence diagrams are provided in the Software Architecture and Design Specification.

7.1 Use-Case Diagram 1 — Inventory and Component Management

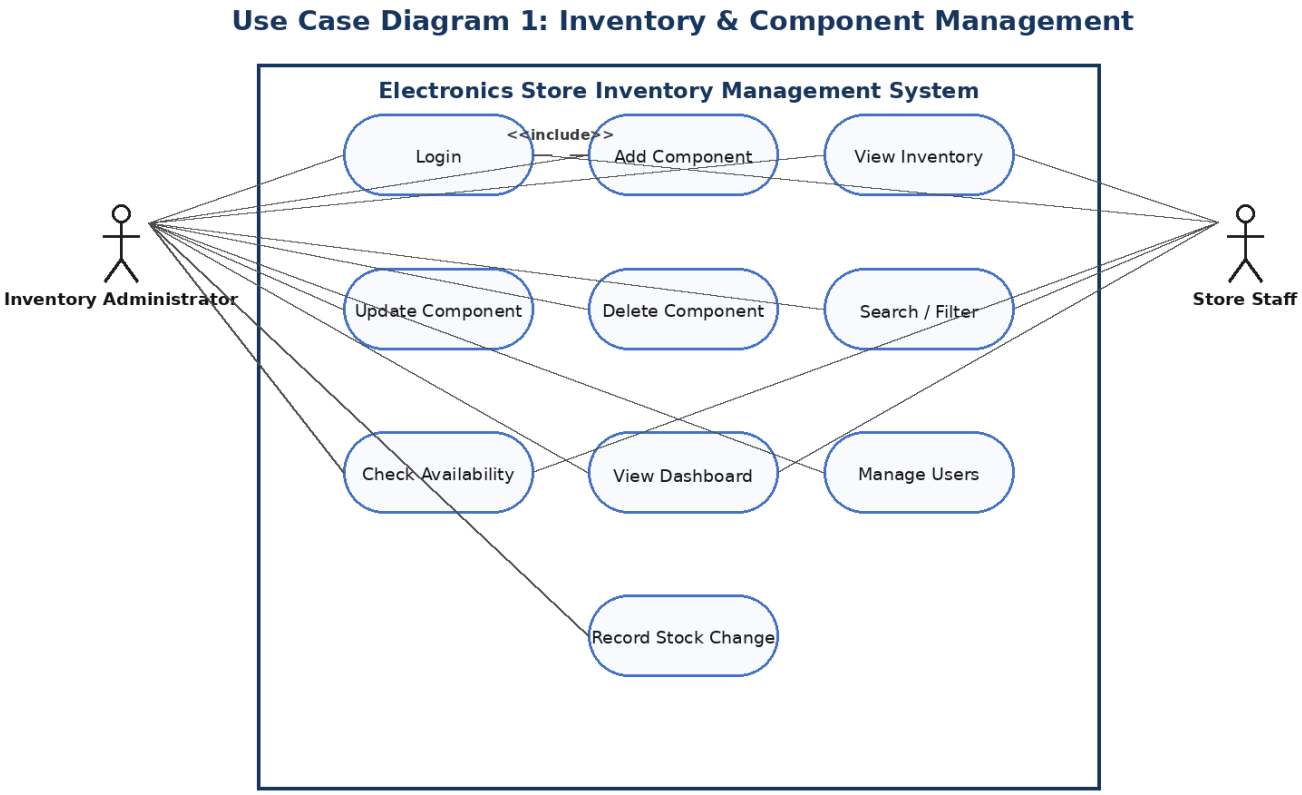


Figure 1 — Inventory and Component Management Use-Case Diagram

7.2 Use-Case Diagram 2 — Reporting, Availability and Monitoring

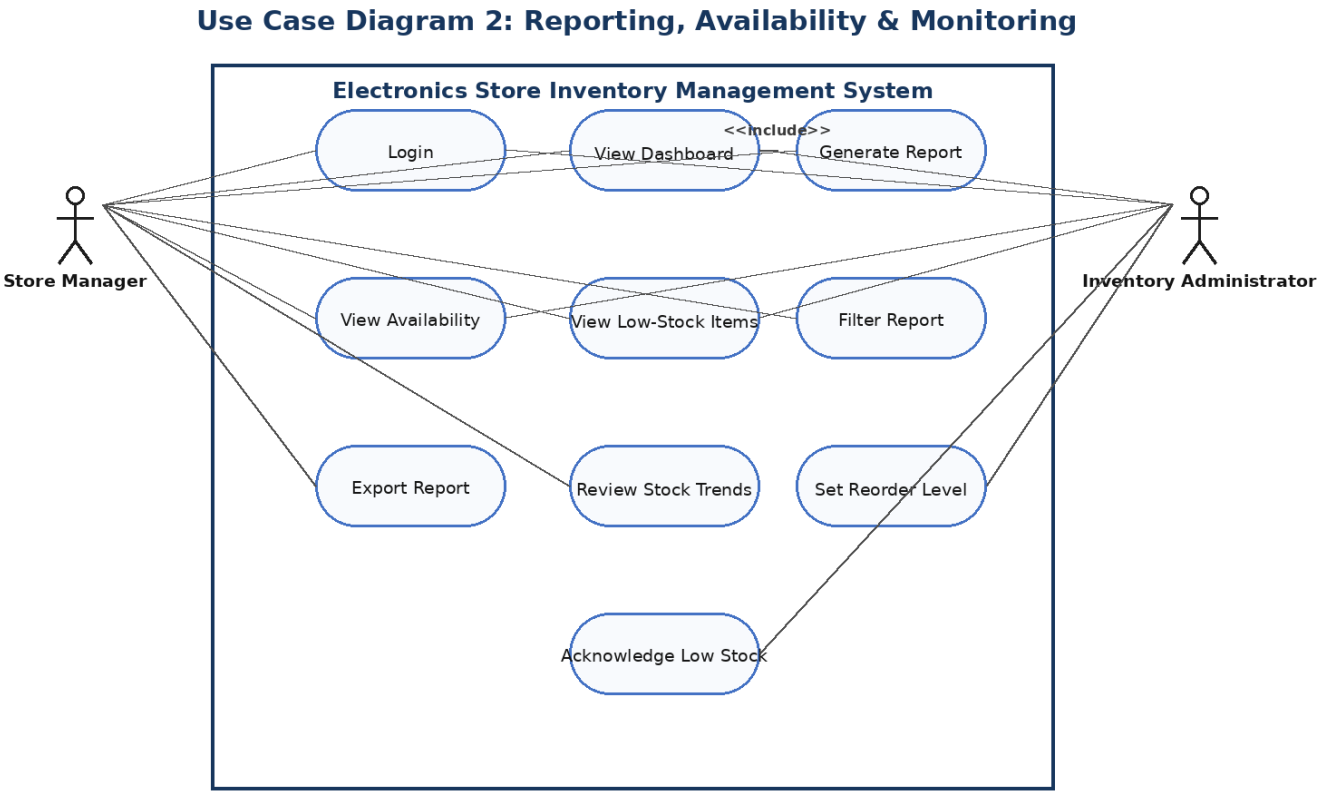


Figure 2 — Reporting, Availability and Monitoring Use-Case Diagram

8. Requirements Traceability Matrix (RTM)

Part 1 of 3. Status N means Not tested in the Phase 1 baseline; later test execution evidence is recorded in Phase 3.

Req ID	Requirement (short)	Section / design spec	Module	Test case(s)	Status	Comments
ESIM-F-001	User Authentication	4.1 / DS-01	AuthModule	TC-AUTH-01	N	
ESIM-F-002	Logout and Session Management	4.2 / DS-02	SessionManager	TC-AUTH-02	N	
ESIM-F-003	Add Component	4.3 / DS-03	InventoryModule	TC-INV-01	N	
ESIM-F-004	View Inventory	4.4 / DS-04	InventoryView	TC-INV-02	N	
ESIM-F-005	Search / Filter	4.5 / DS-05	SearchFilterModule	TC-SEARCH-01	N	
ESIM-F-006	Update Component	4.6 / DS-06	InventoryModule	TC-INV-03	N	
ESIM-F-007	Delete Component	4.7 / DS-07	InventoryModule	TC-INV-04	N	
ESIM-F-008	Availability Status	4.8 / DS-08	AvailabilityService	TC-AVAIL-01	N	
ESIM-F-009	Low-Stock Identification	4.9 / DS-09	StockAlertService	TC-LOW-01	N	
ESIM-F-010	Dashboard	4.10 / DS-10	DashboardModule	TC-DASH-01	N	

8. Requirements Traceability Matrix (RTM)

Part 2 of 3. Status N means Not tested in the Phase 1 baseline; later test execution evidence is recorded in Phase 3.

Req ID	Requirement (short)	Section / design spec	Module	Test case(s)	Status	Comments
ESIM-F-011	Inventory Reports	4.11 / DS-11	ReportModule	TC-REP-01	N	
ESIM-F-012	Input Validation	4.12 / DS-12	ValidationModule	TC-VAL-01	N	
ESIM-F-013	Persistent Storage	4.13 / DS-13	Django ORM / Database	TC-DATA-01	N	
ESIM-F-014	Authorization	4.14 / DS-14	AuthorizationModule	TC-SEC-01	N	
ESIM-F-015	Error Handling	4.15 / DS-15	ErrorHandler	TC-ERR-01	N	
ESIM-F-016	Inventory Change Tracking	4.16 / DS-16	AuditLogModule	TC-AUD-01	N	
ESIM-NF-001	90% of normal-load inventory list, search and CRUD requests within 2 s	5 / DS-NF-01	NFRModule	TC-PERF-01	N	
ESIM-NF-002	Support at least 1,000 component records without functional failure	5 / DS-NF-02	NFRModule	TC-PERF-02	N	
ESIM-NF-003	Transactions complete or preserve the prior valid record	5 / DS-NF-03	NFRModule	TC-REL-01	N	
ESIM-NF-004	At least 99% availability during the controlled test window	5 / DS-NF-04	NFRModule	TC-REL-02	N	

8. Requirements Traceability Matrix (RTM)

Part 3 of 3. Status N means Not tested in the Phase 1 baseline; later test execution evidence is recorded in Phase 3.

Req ID	Requirement (short)	Section / design spec	Module	Test case(s)	Status	Comments
ESIM-NF-005	New user completes common tasks without developer assistance	5 / DS-NF-05	NFRModule	TC-UX-01	N	
ESIM-NF-006	Separate presentation, business, security and data-access modules	5 / DS-NF-06	NFRModule	Code review	N	
ESIM-NF-007	Correct behavior in current Chrome, Edge and Firefox	5 / DS-NF-07	NFRModule	TC-COMP-01	N	
ESIM-NF-008	Restore a project inventory backup in the defined environment	5 / DS-NF-08	NFRModule	TC-REC-01	N	
ESIM-SR-001	Protected pages require authentication	5.1.2 / DS-SR-01	SecurityModule	TC-SEC-01	N	
ESIM-SR-002	Authorize inventory and user-management operations	5.1.2 / DS-SR-02	SecurityModule	TC-SEC-02	N	
ESIM-SR-003	Do not store passwords in plaintext	5.1.2 / DS-SR-03	SecurityModule	TC-SEC-03	N	
ESIM-SR-004	Protect applicable state-changing requests with CSRF	5.1.2 / DS-SR-04	SecurityModule	TC-SEC-04	N	
ESIM-SR-005	Validate server input before database writes	5.1.2 / DS-SR-05	SecurityModule	TC-SEC-05	N	
ESIM-SR-006	Do not log passwords or authentication secrets	5.1.2 / DS-SR-06	SecurityModule	TC-SEC-06	N	

9. Validation / Verification Requirements (VFRs)

VFR ID	Mapped Requirement	Measurable Validation Criterion
ESIM-VFR-001	ESIM-F-001	Valid credentials authenticate; invalid credentials do not create an authorized session.
ESIM-VFR-002	ESIM-F-002	After logout, protected inventory operations are denied until re-authentication.
ESIM-VFR-003	ESIM-F-003	A valid component form submission creates exactly one record with the entered values.
ESIM-VFR-004	ESIM-F-004	Every component in the test dataset is retrievable and displayed with stored details.
ESIM-VFR-005	ESIM-F-005	Search/filter results contain only records satisfying the selected criteria.
ESIM-VFR-006	ESIM-F-006	Updating a component changes the selected record without creating a duplicate ID.
ESIM-VFR-007	ESIM-F-007	Confirmed deletion/deactivation removes the selected component from active inventory.
ESIM-VFR-008	ESIM-F-008	Availability matches the documented rule for quantities below, equal to and above the threshold.
ESIM-VFR-009	ESIM-F-009	All components at or below minimum stock are identified.
ESIM-VFR-010	ESIM-F-010	Dashboard totals equal the corresponding counts calculated from the test dataset.
ESIM-VFR-011	ESIM-F-011	Generated report values match relevant stored component and stock records.
ESIM-VFR-012	ESIM-F-012	Missing, duplicate and negative/invalid values are rejected with a clear validation message.
ESIM-VFR-013	ESIM-F-013	A valid component record remains retrievable after logout and subsequent login.
ESIM-VFR-014	ESIM-F-014	A user without required permission is blocked from restricted inventory changes.
ESIM-VFR-015	ESIM-F-015	Failed/invalid operations provide understandable feedback without corrupting valid records.
ESIM-VFR-016	ESIM-F-016	Where audit tracking is enabled, a supported inventory change creates the defined audit information.

10. Phase 1 Validation Checklist

Required Phase 1 Deliverable	Status / Location
Problem statement analysis / feasibility testing	Completed in Sections 1 and 2
SRS list	Functional, non-functional and security requirements included
FR and VFR categorization	FRs and measurable VFRs included
RTM	All defined FRs, NFRs and security requirements traced to tests
Actors identified	Inventory Administrator, Store Staff, Store Manager, QA/Evaluator
Use cases grouped by actors	Covered in Section 7
Use Case Diagrams	Two diagrams included
Technology stack	Python (Django)
Validation focus	Acceptance criteria and planned test references included

11. Assumptions, Risks and Mitigation

Category	Details
Assumption	Component IDs are unique and inventory quantities are maintained through authorized system operations.
Risk	Incorrect stock quantities may lead to incorrect availability or reports.
Risk	Unauthorized users may attempt inventory changes.
Risk	Data loss may occur if database backups are not maintained.
Mitigation	Use validation, authorization, transaction-safe updates, backups and RTM-based testing.

Appendix A — Measurable Acceptance Criteria

This appendix makes the acceptance measures referenced by Sections 4–6 explicit. These are planned criteria; no unperformed manual or operational test is represented as passed.

ID	Measure	Acceptance criterion
ESIM-F-010	Dashboard measures	For a fixed fixture, Active Components = count of rows with is_active=True; Available Units = sum(quantity) for active rows; Low Stock = count of active rows with quantity ≤ minimum_stock. Each displayed value shall equal an independent query over the fixture. Inactive records are excluded from these operational metrics.
ESIM-F-016	Change tracking	Each successful component create, update, and deactivation shall create exactly one ChangeLog row containing component ID, component name, action, actor when authenticated, and timestamp. A rejected operation shall not create a success log.
ESIM-NF-001	Performance	Measure 10 requests for each of list, search, add, edit, and deactivate (50 total) with the Django test client on the stated test environment. At least 45/50 shall complete in under 2.0 seconds.
ESIM-NF-002	Scalability	With at least 1,000 component rows, list and search shall return the correct results, and add, edit, and deactivate shall complete without data corruption.
ESIM-NF-003	Transaction reliability	For each tested write, a successful transaction shall persist all intended changes; an injected failure shall leave the affected component and audit rows at their prior valid state.
ESIM-NF-004	Availability	Run one scheduled 12-hour measurement with one reachability sample every 60 seconds. After excluding predeclared maintenance, measured availability shall be ≥99.00% (eligible downtime ≤1% of eligible time). Save the append-only samples and generated report with timezone, endpoint, start/end, and exclusions.
ESIM-NF-005	Usability	A first-time, non-developer participant shall attempt the five tasks in the usability protocol without coaching. Pass criterion: at least 4/5 tasks completed without assistance and no critical data-loss/security concern. Record task completion, time, attempts, help requests, and participant feedback with consent.
ESIM-NF-006	Modularity	A documented review shall confirm the responsibilities of accounts, inventory, reports, and audit; one-directional imports; and no circular runtime imports among these apps.
ESIM-NF-007	Browser compatibility	On the test date, record exact versions of the then-current stable Chrome, Edge, and Firefox. All ten workflows listed in the browser compatibility report shall pass in each browser; mark a browser pending until live execution evidence exists.
ESIM-NF-008	Recoverability	Restore a backup to a new isolated destination. Require database integrity_check=ok and exact equality of user-table names and rows between source and restored copies; preserve the JSON result and never overwrite the live database.

Interpretation note for ESIM-F-010: The SRS now labels the implemented active-only metric explicitly. This avoids treating deactivated historical rows as active inventory. If the requirement owner intends a separate all-record total, that is a product decision and requires a separate metric and approved change.

Planned standards and document basis

The section organization follows the course-provided SRS template and the IEEE 830 SRS outline, with IEEE/ISO requirements-engineering guidance. The supplied template remains the course-specific layout authority; this project does not claim formal standards certification.

References: IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications (superseded by ISO/IEC/IEEE 29148); ISO/IEC/IEEE 29148 requirements engineering. See the IEEE Standards Association references in the team submission materials.

> **Normative clarification for ESIM-F-016**
>
> This mandatory criterion supersedes the conditional wording in Section 4.16 and VFR-016:
> each successful create, update, or deactivation shall write one complete ChangeLog record; rejected writes shall not.